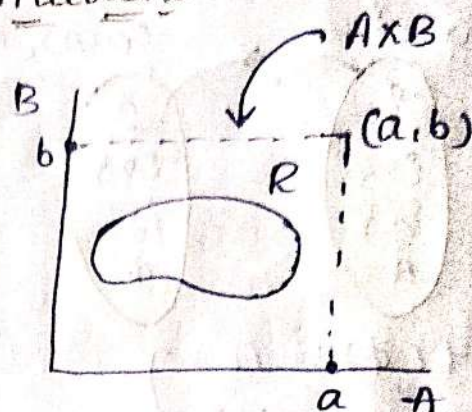


Relations and their Representations

The Cartesian product of two sets  $A$  and  $B$  is denoted by  $A \times B$  and defined by

$$A \times B = \{a, b\} : a \in A, b \in B\}$$



Ex:-

$$A = \{a, b\}, B = \{0, 1\}$$

$$A \times B = \{a, b\} \times \{0, 1\} \\ = \{(a, 0), (a, 1), (b, 0), (b, 1)\}$$

⇒ Every Subset in  $A \times B$  is called, Relation  $R$  from  $A$  to  $B$ .

Ex:-

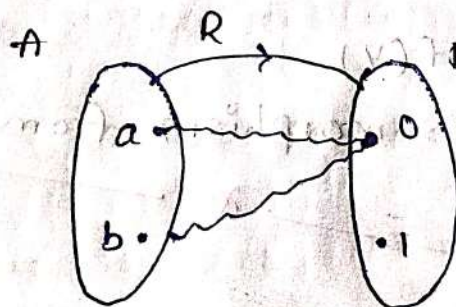
$$A = \{a, b\}, B = \{0, 1\}$$

$$A \times B = \{(a, 0), (a, 1), (b, 0), (b, 1)\}$$

∴ No of subsets  
=  $2^4 = 16$

Taking a Subset in  $A \times B$

$R = \{(a, 0), (b, 0)\}$  is a relation from  $A$  to  $B$ .

Properties of the relation

(i) Let  $R: A \rightarrow B$  be a relation, then

$$\boxed{R \subseteq A \times B}$$

(ii) Let  $R: A \rightarrow B$  be a relation, then

3. Let  $A, B$  belongs to  $\mathcal{P}(S)$  (2)
- (a, b)  $\in R$  then
- (i)  $a$  is related to  $b$
  - (ii)  $a$  is associated to  $b$
  - (iii)  $aRb$
  - (iv)  $a \sim b$

4.  $|A| = m$  and  $|B| = n$

No of relations = No of subsets in  $A \times B$

No of relations from  $A$  to  $B$  is  $2^{|A \times B|} = 2^{|A| \cdot |B|} = 2^{mn}$

No of relations from  $A$  to  $B$  is

$$2^{|A \times B|} = 2^{|A| \cdot |B|} = 2^{mn}$$

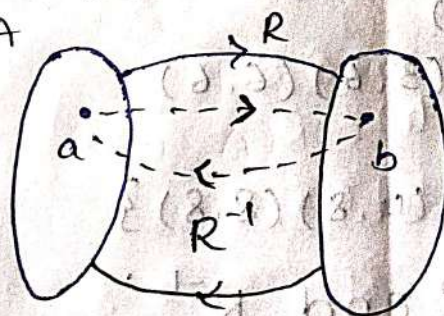
No of relations from  $B$  to  $B$  is  $2^{|B \times B|} = 2^{|B| \cdot |B|} = 2^{|B|^2}$

No of relations from  $B$  to  $A$  is

$$2^{|B \times A|} = 2^{|B| \cdot |A|} = 2^{nm}$$

5. Let  $R: A \rightarrow B$  be a relation, then the inverse relation of  $R$  is denoted by  $R^{-1}: B \rightarrow A$  and defined by

$$R^{-1} = \{(b, a) \in B \times A : (a, b) \in A \times B\}$$



Sol. The given set is

$$\therefore A = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

and the given relation is

$$\boxed{a R b \Leftrightarrow \frac{a}{b} \in Z} \rightarrow \textcircled{1}$$

$\forall a, b \in A$ , where  $Z = \{0, \pm 1, \pm 2, \dots\}$

Now to Construct  $R$  from  $\textcircled{1}$  follows.

$$R = \{(1,1), (1,2), (1,3), \dots, (8,8)\}$$

$$\Rightarrow R = \{(1,1)$$

$$(2,1), (2,2)$$

$$(3,1), (3,3)$$

$$(4,1), (4,2), (4,4)$$

$$(5,1), (5,5)$$

$$(6,1), (6,2), (6,3), (6,6)$$

$$(7,1), (7,7)$$

$$(8,1), (8,2), (8,4), (8,8)\}$$

and its inverse relation is

$$R^{-1} = \{(1,1)$$

$$(1,2), (2,2)$$

$$(1,3), (3,3)$$

$$(1,4), (2,4), (4,4)$$

$$(1,5), (5,5)$$

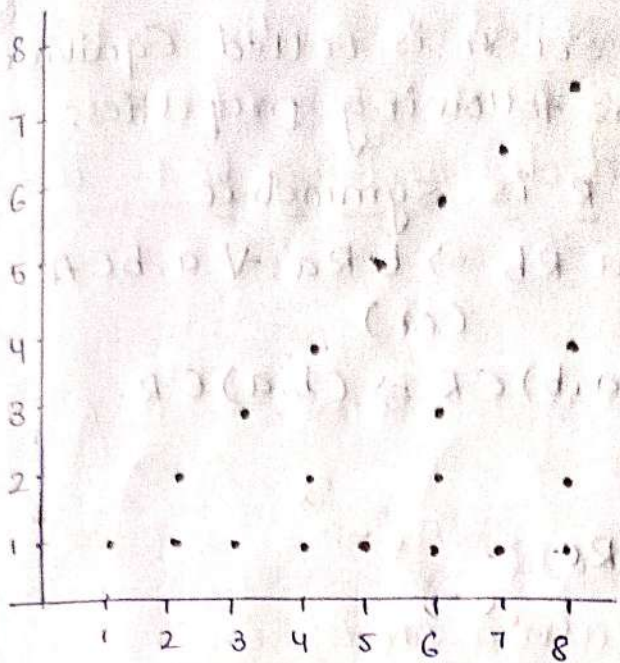
$$(1,6), (2,6), (3,6), (6,6)$$

$$(1,7), (7,7)$$

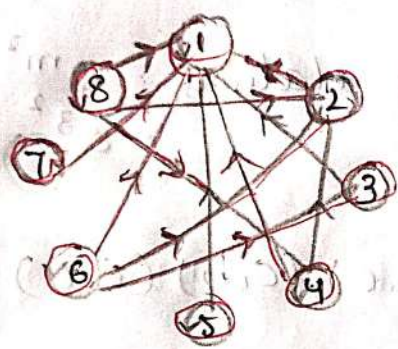
$$(1,8), (2,8), (4,8), (8,8)\}$$

Representation of  $R$  and  $R^{-1}$

(a) Grid Representation of  $R$  (a) Grid representation of  $R^{-1}$

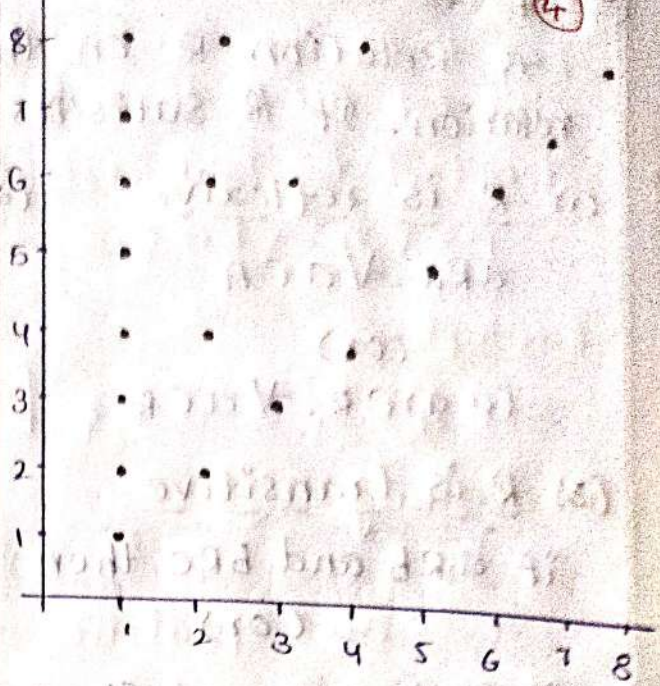


(b) Directed graph of  $R$

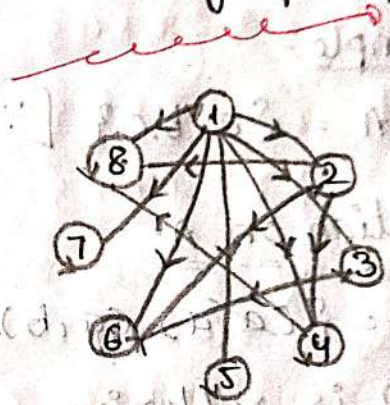


(c) Binary matrix of  $R$

|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 4 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| 5 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 |
| 7 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 8 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 |



(b) Directed graph of  $R^{-1}$



(c) Binary matrix of  $R^{-1}$

|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 3 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 5 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 6 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 7 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 8 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

Question-1 Completed.

# Equivalence Relations

A Relation  $R$  on the Set  $A$  is called Equivalence relation. If  $R$  satisfies the following properties:

- (1)  $R$  is reflexive  
 $aRa, \forall a \in A$   
 (or)  
 $(a,a) \in R, \forall a \in R$
- (2)  $R$  is Symmetric  
 $aRb \Rightarrow bRa, \forall a, b \in A$   
 (or)  
 $(a,b) \in R \Rightarrow (b,a) \in R$

- (3)  $R$  is transitive  
 if  $aRb$  and  $bRc$ , then  $aRc$   
 (or)

if  $(a,b) \in R$  and  $(b,c) \in R$ , then  $(a,c) \in R$

Example:-

Set  $A = \{a, b, c\}$ .  $\therefore$  Total no of relations =  $2^{m^2}$   
 $= 2^{3^2} = 2^9$

Relation on  $A$

$$R = \{(a,a), (a,b), (b,a), (b,c), (c,b), (c,c), (b,b)\}$$

- (i)  $R$  is reflexive  
 $xRx, \forall x \in A$   
 $\therefore R$  is reflexive
- (ii)  $R$  is Symmetric  
 $(a,b) \in R \Rightarrow (b,a) \in R$   
 $(b,c) \in R \Rightarrow (c,b) \in R$   
 $\therefore R$  is Symmetric

- (iii)  $R$  is transitive

$$(a,a), (a,b) \in R \Rightarrow (a,b) \in R$$

$$(a,b), (b,a) \in R \Rightarrow (a,b) \in R$$

$$(b,c), (c,b) \in R \Rightarrow (b,b) \in R$$

$\therefore R$  is equivalence relation on  $A$

- (2) a) The given set is  $Z^+ = N = \{1, 2, 3, \dots\}$  and the relation is

$$nRm \Leftrightarrow n|m \rightarrow \textcircled{1}$$

$$\forall n, m \in Z^+$$

- (i)  $R$  is reflexive

We know that every positive integer  $n$  is divisible by itself.

(i.e.)  $\forall n, \forall m \in \mathbb{Z}^+$   
 $\Rightarrow n R m, \forall n \in \mathbb{Z}^+$  by ①

$\therefore R$  is reflexive.

(ii)  $R$  is Symmetric

Suppose  $n R m$  then by ①

$$n/m$$

$$\Rightarrow m/n$$

$$\Rightarrow m R n$$

Ex:  $3/12$  but  $12/3$   $\therefore R$  is not Symmetric

(iii)  $R$  is transitive

Suppose  $n R m$  and  $m R l$ . Then by ①

$$n/m \text{ and } m/l$$

$$\Rightarrow n/l$$

$$\Rightarrow n R l$$

$\therefore R$  is transitive.

Hence, the relation  $R$  defined by ① is not equivalence on  $\mathbb{Z}^+$ .

(b) The given set is  $\mathbb{R}$  (real numbers) and the relation is

$$x R y \Leftrightarrow x^2 = y^2 \rightarrow ①$$

$$\forall x, y \in \mathbb{R}$$

(i)  $R$  is reflexive

We know that

$$x^2 = x^2, \forall x \in \mathbb{R}$$

$$\Rightarrow x R x, \forall x \in \mathbb{R}$$

(ii)  $R$  is Symmetric

Suppose  $x R y$  then by ①

$$x^2 = y^2 \Rightarrow y^2 = x^2$$

$$\Rightarrow y R x, \therefore R \text{ is Symmetric}$$

(iii)  $R$  is transitive

Suppose  $x R y$  and  $y R z$ . then by ①

$$x^2 = y^2 \text{ and } y^2 = z^2$$

$$\Rightarrow x^2 = z^2$$

$$\Rightarrow x R z$$

Hence, the relation  $R$  defined by ① is equivalence

on  $\mathbb{R}$  (set  $\mathbb{R}$ ) (Question - 2 Completed)

③ Equivalence Class  
 Let  $R$  be the equivalence relation on the set  $A$ .  
 Let  $a \in A$  then the equivalence class of  $a$ , is  
 denoted by  $[a]$  or  $\bar{a}$  and defined by  $[a] = \bar{a} =$   
 $\{x \in A : x R a\}$

Example:

Set  $A = \{a, b, c\}$  [Total relations:  $2^{3^2} = 2^9$ ]

(i) Taking

$$R_1 = \{(a, a), (b, b), (c, c)\}$$

It is an equivalence relation on  $A$ .

Classes of  $A$

$$[a] = \{x : x R a\} = \{a\} \text{ --- ①}$$

$$[b] = \{x : x R b\} = \{b\} \text{ --- classes. ②}$$

$$[c] = \{x : x R c\} = \{c\} \text{ --- ③}$$

(ii) Taking

$$R_2 = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$$

It is an equivalence relation on  $A$ .

Classes of  $A$

$$[a] = \{x : x R a\} = \{a, b, c\}$$

$$[b] = \{x : x R b\} = \{a, b, c\}$$

$$[c] = \{x : x R c\} = \{a, b, c\}$$

(iii) Taking

$$R_3 = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$$

It is an equivalence relation on  $A$ .

Classes of  $A$

$$[a] = \{x : x R a\} = \{a, b\}$$

$$[b] = \{x : x R b\} = \{a, b\}$$

$$[c] = \{x : x R c\} = \{c\}$$

Q. The given set is  $Z = \{0, \pm 1, \pm 2, \dots\}$  and the relation on  $Z$  is

$$n R m \Leftrightarrow |n| = |m| \rightarrow \textcircled{1}$$

if  $n, m \in Z$ .

(i)  $|n| = |n|, \forall n \in Z$

$\Rightarrow n R n, \forall n \in Z$

(ii)  $n R m \Rightarrow |n| = |m|$

$\Rightarrow |m| = |n|$

$\Rightarrow m R n$

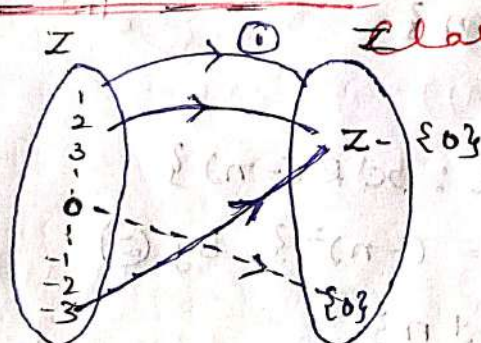
(iii)  $n R m \Rightarrow |n| = |m|$  and  $|m| = |l|$

$\Rightarrow |m| = |l|$

$\Rightarrow m R l$

$\therefore$  the relation  $\textcircled{1}$  is equivalence <sup>relation</sup> on  $Z$ .

Classes of  $Z$  w.r. to  $\textcircled{1}$



$$\begin{aligned} [0] &= \bar{0} = \{x : x R 0\} = \{x : |x| = |0|\}, \text{ by } \textcircled{1} \\ &= \{x : x = 0\} \\ &= \{0\} \end{aligned}$$

$$\begin{aligned} [n] &= \bar{n} = \{x : x R n\} = \{x : |x| = |n|\}, \text{ by } \textcircled{1} \\ &= \{x : x = \pm n\} \\ &= Z - \{0\} \end{aligned}$$

Similarly

$$[-n] = Z - \{0\}$$

Q. (b) The given set is  $Z = \{0, \pm 1, \pm 2, \dots\}$  and the relation is

$$n R m \Leftrightarrow n^2 = m^2 \rightarrow \textcircled{1}$$

$$\forall n, m \in Z$$

(i) Since  $n^2 = n^2, \forall n \in Z$

$\Rightarrow n R n, \forall n \in Z$ , by  $\textcircled{1}$

(i)  $nRm \Rightarrow n^2 = m^2$  by (1)

$\Rightarrow m^2 = n^2$

$\Rightarrow mRn$

(ii)  $nRm$  and  $mRl \Rightarrow n^2 = m^2, m^2 = l^2$

$\Rightarrow n^2 = l^2$

$\Rightarrow nRl$  by (1)

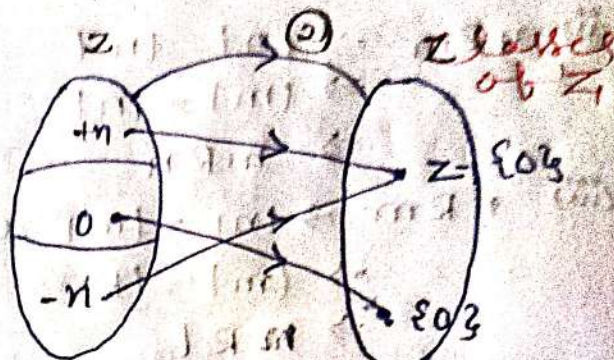
Hence, the relation (1) is an equivalence relation on  $\mathbb{Z}$ .

Equivalence Classes

$[0] = \bar{0} = \{x \in \mathbb{Z} : xR0\}$

$= \{x \in \mathbb{Z} : x^2 = 0^2\}$  by (2)

$= \{0\}$



$[n] = \bar{n} = \{x \in \mathbb{Z} : xRn\}$  where  $n \neq 0$

$\Rightarrow \{x \in \mathbb{Z} : x^2 = n^2\}$  by (2)

$\Rightarrow \{x \in \mathbb{Z} : x = \pm n\}$

$\Rightarrow \mathbb{Z} - \{0\}$

$[-n] = -\bar{n} = \{x \in \mathbb{Z} : xR(-n)\}$

$= \{x \in \mathbb{Z} : x^2 = (-n)^2\}$  by (2)

$= \{x \in \mathbb{Z} : x = \pm n\}$

(iii)  $\mathbb{Z} - \{0\} = \mathbb{Z} - \{0\} : x \neq 0$  Question-3 Completed

Example-4

Show that (a) the relation defined on  $\mathbb{Q}$  by  $xRy \Leftrightarrow (x-y) \in \mathbb{Z}$  is an equivalence relation, and find the equivalence classes containing 2 and  $\frac{1}{4}$ , and (b) the relation  $R$  on  $\mathbb{Z}^+ \times \mathbb{Z}^+$  defined by  $(m,n)R(p,q) \Leftrightarrow m+q = n+p$  is an equivalence relation and find the equivalence classes of  $(2,1)$  and  $(1,3)$ .

Sol: (a) The given set is  $\mathbb{Q}$  (Set of rationals) and the relation is  $xRy \Leftrightarrow (x-y) \in \mathbb{Z} \rightarrow (1)$

$\forall x, y \in \mathbb{Q}$

(i) Since  $0 \in \mathbb{Z} \Rightarrow x-x \in \mathbb{Z}, \forall x \in \mathbb{Q}$

$\Rightarrow xRx, \forall x \in \mathbb{Q}$  by (1)

$$\begin{aligned}
 \text{(ii)} \quad x R y &\Rightarrow x - y \in \mathbb{Z} \text{ by } \textcircled{1} \\
 &\Rightarrow -(y - x) \in \mathbb{Z} \\
 &\Rightarrow -(-(-y - x)) \in \mathbb{Z} \\
 &\Rightarrow (-y - x) \in \mathbb{Z} \\
 &\Rightarrow y R x, \forall y, x \in \mathbb{Q}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad x R y \text{ and } y R z &\Rightarrow (x - y) \in \mathbb{Z} \text{ and } (y - z) \in \mathbb{Z} \text{ by } \textcircled{1} \\
 &\Rightarrow (x - y) + (y - z) \in \mathbb{Z} \\
 &\Rightarrow (x - z) \in \mathbb{Z} \\
 &\Rightarrow x R z, \forall x, z \in \mathbb{Q} \text{ by } \textcircled{1}
 \end{aligned}$$

$\therefore$  The relation  $\textcircled{1}$  is an equivalence relation on  $\mathbb{Q}$ .

Equivalence classes

$$[2] = \bar{2} = \{x \in \mathbb{Q} : x R 2\}$$

$$\textcircled{2} = \{x \in \mathbb{Q} : (x - 2) \in \mathbb{Z}\} \text{ by } \textcircled{1}$$

$$= \{x \in \mathbb{Q} : x - 2 = y, y \in \mathbb{Z}\}$$

$$= \{x \in \mathbb{Q} : x = y + 2, y \in \mathbb{Z}\}$$

$$= \{y + 2 : y \in \mathbb{Z}\} = \mathbb{Z} + 2$$

$$\left[\frac{1}{4}\right] = \bar{\frac{1}{4}} = \{x \in \mathbb{Q} : x R \frac{1}{4}\}$$

$$= \{x \in \mathbb{Q} : (x - \frac{1}{4}) \in \mathbb{Z}\} \text{ by } \textcircled{1}$$

$$= \{x \in \mathbb{Q} : (x - \frac{1}{4}) = y, y \in \mathbb{Z}\}$$

$$= \{x \in \mathbb{Q} : x = y + \frac{1}{4}, y \in \mathbb{Z}\}$$

$$= \{y + \frac{1}{4} : y \in \mathbb{Z}\} = \mathbb{Z} + \frac{1}{4}$$

$\textcircled{16}$  The given set is

$$\mathbb{Z}^+ \times \mathbb{Z}^+ = \{(m, n) : m \in \mathbb{Z}^+, n \in \mathbb{Z}^+\} \text{ and}$$

the relation is

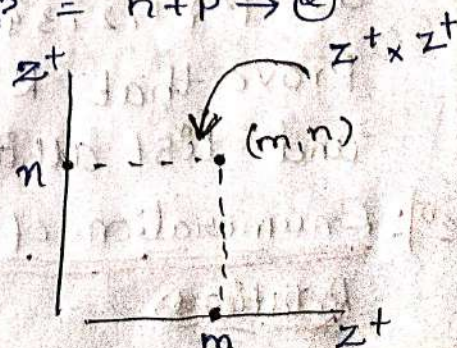
$$(m, n) R (p, q) \Leftrightarrow m + q = n + p = n + p \rightarrow \textcircled{2}$$

$$\forall (m, n), (p, q) \in \mathbb{Z}^+ \times \mathbb{Z}^+$$

Transitive

$$\text{(i) Since } m + n = n + m, \forall (m, n).$$

$$\Rightarrow (m, n) R (m, n) \text{ by } \textcircled{1}$$



$$(ii) (m, n) R (p, q) \Rightarrow m + q = n + p \text{ by } (i) \\ \Rightarrow n + p = m + q \\ \Rightarrow (p, q) R (m, n)$$

$$(iii) (m, n) R (p, q) \text{ and } (p, q) R (x, y)$$

$$\Rightarrow n + q = n + p \text{ and } p + y = x + q \text{ by } (i)$$

$$\Rightarrow m + q - p + y = (n + p) - (x + q)$$

$$\Rightarrow m + q = x + n$$

$$\Rightarrow (m, n) R (x, y)$$

$\therefore$  The relation (i) is an equivalence relation on

$$\mathbb{Z}^+ \times \mathbb{Z}^+$$

Equivalence classes

$$[(2, 1)] = (\overline{2, 1}) = \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : (x, y) R (2, 1)\}$$

$$= \{(x, y) : x + 1 = 2 + y\} \text{ by } (i)$$

$$= \{(x, y) : x = y + 1\}$$

$$= \{(y + 1, y) : y \in \mathbb{Z}^+\}$$

$$= \{(\mathbb{Z}^+ + 1, \mathbb{Z}^+)\}$$

$$[(1, 3)] = (\overline{1, 3}) = \{(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+ : (x, y) R (1, 3)\}$$

$$= \{(x, y) : x + 3 = 1 + y\} \text{ by } (i)$$

$$= \{(x, y) : x = y - 2\}$$

$$= \{(y - 2, y) : y \in \mathbb{Z}^+\}$$

$$= \{(\mathbb{Z}^+ - 2, \mathbb{Z}^+)\}$$

Example-5

Find the number of all equivalence relations on the sets (i)  $\{a, b\}$ , (ii)  $\{a, b, c\}$  and (iii)

$\{a, b, c, d\}$ . Construct a relation  $R$  on power

Set  $P = \{1, 2, 3\}$  by the rule  $A R B \Leftrightarrow |A| = |B|$ .

Prove that  $R$  is an equivalence relation and list all its equivalence classes.

Sol: Enumeration of Equivalence Relations:

Relations

(i) The given set is

$$A = \{a, b\}$$

Total no of relations on  $A$  is

$$2^{m^2} = 2^{2^2} = 2^4 = 16$$

Among these  $2^4$  ~~relations~~ <sup>relations</sup>, we have

$$R_1 = \{(a, a), (b, b)\} \text{ [Identity relation]}$$

$$R_2 = \{(a, a), (b, b), (a, b), (b, a)\} \text{ [Universal Relation]}$$

Are equivalence relation on  $A = \{a, b\}$

(ii) The given set

$$A = \{a, b, c\}$$

$$\text{Total no of relations} = 2^{m^2} = 2^{3^2} \Rightarrow 2^9$$

Among these  $2^9$  ~~the~~ <sup>relations</sup>, we have

$$R_1 = \{(a, a), (b, b), (c, c)\} \text{ (Identity)}$$

$$R_2 = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$$

$$R_3 = \{(a, a), (b, b), (c, c), (b, c), (c, a)\}$$

$$R_4 = \{(a, a), (b, b), (c, c), (c, a), (a, c)\}$$

$$R_5 = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c)$$

$$(c, a), (c, b), (c, c)\} \text{ (Universal)}$$

Are all equivalence relations on  $A = \{a, b, c\}$

(iii) The given set is

$$A = \{a, b, c, d\}$$

$$\text{Total no of } \text{~~notifications~~ <sup>relations</sup>} = 2^{4^2} = 2^{16}$$

Among these  $2^{16}$  ~~notifications~~ <sup>relations</sup>, we have

$$R_1 = \{(a, a), (b, b), (c, c), (d, d)\}$$

$$R_2 = \{(a, a), (b, b), (c, c), (d, d), (a, b), (b, a)\}$$

$$R_3 = \{(a, a), (b, b), (c, c), (d, d), (a, c), (c, a)\}$$

$$R_4 = \{(a, a), (b, b), (c, c), (d, d), (a, d), (d, a)\}$$

$$R_5 = \{(a, a), (b, b), (c, c), (d, d), (b, c), (c, b)\}$$

$$R_6 = \{(a, a), (b, b), (c, c), (d, d), (c, d), (d, c)\}$$

$$R_7 = \{(a, a), (b, b), (c, c), (d, d), (c, d), (d, c)\}$$

$R_9 = \{(a,a), (b,b), (c,c), (d,d), (a,b), (b,a), (c,d), (d,c)\}$   
 $R_{10} = \{(a,a), (b,b), (c,c), (d,d), (a,b), (b,a), (c,d)\}$   
 $R_{11} = \{(a,a), (b,b), (c,c), (d,d), R, UR_3, UR_5\}$   
 $R_{12} = \{(a,a), (b,b), (c,c), (d,d), R, UR_4, UR_6\}$   
 $R_{13} = R_5 \cup R_7 \cup R_8$   
 $R_{14} = \{(a,a), (b,b), (c,c), (d,d), (a,b), (b,a), (c,d), (d,c)\}$   
 $R_{15} = \{(a,a), (b,b), (c,c), (d,d), (a,b), (b,a), (c,b), (b,c)\}$

(13)

are all equivalence relations on  
 $A = \{a, b, c, d\}$

| No of elements<br>in A | No of relations<br>in A | No of Equivalence<br>relations on A. |
|------------------------|-------------------------|--------------------------------------|
| 2                      | $2^{2^2}$               | 2                                    |
| 3                      | $2^{3^2}$               | 5                                    |
| 4                      | $2^{4^2}$               | 15                                   |

$\{(a,b), (b,a), (c,c), (d,d)\}$   
 $\{(a,b), (b,a), (c,d), (d,c)\}$   
 $\{(a,b), (b,a), (c,c), (d,d)\}$   
 $\{(a,b), (b,a), (c,c), (d,d), (a,c), (c,a)\}$   
 $\{(a,b), (b,a), (c,c), (d,d), (a,d), (d,a)\}$   
 $\{(a,b), (b,a), (c,c), (d,d), (a,c), (c,a), (a,d), (d,a)\}$   
 $\{(a,b), (b,a), (c,c), (d,d), (a,c), (c,a), (b,d), (d,b)\}$   
 $\{(a,b), (b,a), (c,c), (d,d), (a,d), (d,a), (b,c), (c,b)\}$   
 $\{(a,b), (b,a), (c,c), (d,d), (a,c), (c,a), (b,d), (d,b), (b,c), (c,b)\}$

The given Set is  $S = \{1, 2, 3\}$  and its powerset is  $P(S) = (\{1, 2, 3\})$

= Set of all Subsets of  $S$

=  $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, S\}$

and the given relation is

$$ARB \Leftrightarrow |A| = |B| \rightarrow \textcircled{1}$$

$$\forall A, B \in P(S)$$

(i)  $R$  is reflexive

Since  $|A| = |A|, \forall A \in P(S)$

$\Rightarrow ARA, \forall A \in P(S)$  by  $\textcircled{1}$

$\Rightarrow R$  is Reflexive.

(ii)  $R$  is Symmetric

$ARB \Rightarrow |A| = |B|$  by  $\textcircled{1}$

$$\Rightarrow |B| = |A|$$

$$\Rightarrow BRA \rightarrow \textcircled{1}$$

(iii)  $R$  is transitive

$ARB$  and  $BRC \Rightarrow |A| = |B|$  and  $|B| = |C|$  by  $\textcircled{1}$

$$\Rightarrow |A| = |C|$$

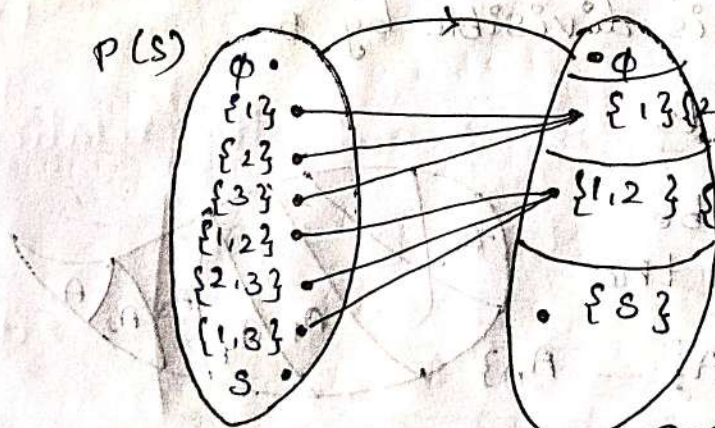
$$\Rightarrow ARC$$

$R$  is transitive

The Relation  $\textcircled{1}$  is an equivalence relation on  $P(S)$ .

Equivalence class

Classes of  $P(S)$



partitions of  $P(S)$

$$\begin{aligned} C(\emptyset) &= \overline{\emptyset} = \{A \in P(S) : A R \emptyset\} \\ &= \{A \in P(S) : |A| = |\emptyset|\} \\ &= \{A \in P(S) : |A| = 0\} \\ &= \{A \in P(S) : A = \emptyset\} = \{\emptyset\} \end{aligned}$$

$$(\{1\}) = \{\bar{1}\} = \{A \in P(S) : A \cap \{1\} = \emptyset\}$$

$$= \{A \in P(S) : |A| \cap |\{1\}| = 0\}$$

$$= \{A \in P(S) : |A| = 1\}$$

$$= \{A \in P(S) : |A| = \{1\}, \{2\}, \{3\}\}$$

$$= \{1\}, \{2\}, \{3\}$$

$$(\{2\}) = \{1\}, \{2\}, \{3\}$$

$$(\{3\}) = \{1\}, \{2\}, \{3\}$$

$$(\{1, 2\}) = \{\bar{1, 2}\} = \{A \in P(S) : A \cap \{1, 2\} = \emptyset\}$$

$$= \{A \in P(S) : |A| \cap |\{1, 2\}| = 0\}$$

$$= \{A \in P(S) : |A| = 2\}$$

$$= \{A \in P(S) : A = \{1, 3\}, \{2, 3\}\}$$

$$= \{1, 3\}, \{2, 3\}$$

$$\{2, 3\} = \{1, 3\} = \{1, 2\}, \{1, 3\}, \{2, 3\}$$

$$[\bar{S}] = \bar{S} = \{A \in P(S) : A \cap S = \emptyset\}$$

$$= \{A \in P(S) : |A| \cap |S| = 0\}$$

$$= \{A \in P(S) : |A| = 3\}$$

$$= \{S\}$$

Imp

Ex Example

Explain the Concept of partitions of a set and find all possible partitions of the sets (i)  $\{a, b\}$  (ii)  $\{a, b, c\}$  and (iii)  $\{a, b, c, d\}$ . partition the structure  $Z_5$  of all equivalence classes of the equivalence relation  $aRb$  if and only if  $a-b$  is divisible by 5.

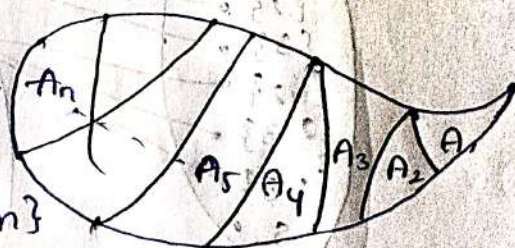
partitions of the sets

Let  $A_1, A_2, \dots, A_n$  be

non empty set then the

Collection  $P = \{A_1, A_2, \dots, A_n\}$

is called partition of  $A$ .



if (i)  $A : \cap A_j = \emptyset$  for all  $i \neq j$

(ii)  $\bigcup_{i=1}^n A_i = A$

The given set is  $A = \{a, b\}$

$\therefore$  The partitions of  $A = \{a, b\}$  are

$$P_1 = \{\{a\}, \{b\}\}$$

$$P_2 = \{\{a, b\}\}$$

The given set is  $A = \{a, b, c\}$

$\therefore$  The partitions of  $A = \{a, b, c\}$  are

$$P_1 = \{\{a\}, \{b\}, \{c\}\}$$

$$P_2 = \{\{a\}, \{b, c\}\}$$

$$P_3 = \{\{a, b\}, \{c\}\}$$

$$P_4 = \{\{a, c\}, \{b\}\}$$

$$P_5 = \{\{a, b, c\}\}$$

The given set is

$$A = \{a, b, c, d\}$$

$\therefore$  The partitions of  $A$  are

$$P_1 = \{\{a\}, \{b\}, \{c\}, \{d\}\}$$

$$P_2 = \{\{a\}, \{b, c, d\}\}$$

$$P_3 = \{\{b\}, \{a, c, d\}\}$$

$$P_4 = \{\{c\}, \{a, b, d\}\}$$

$$P_5 = \{\{d\}, \{a, b, c\}\}$$

$$P_6 = \{\{a, b\}, \{c, d\}\}$$

$$P_7 = \{\{a, c\}, \{b, d\}\}$$

$$P_8 = \{\{a, d\}, \{b, c\}\}$$

$$P_9 = \{\{a\}, \{b\}, \{c, d\}\}$$

$$P_{10} = \{ \{a\}, \{c\}, \{b, d\} \}$$

$$P_{11} = \{ \{a\}, \{d\}, \{b, c\} \}$$

$$P_{12} = \{ \{b\}, \{c\}, \{a, d\} \}$$

$$P_{13} = \{ \{c\}, \{d\}, \{a, b\} \}$$

$$P_{14} = \{ \{b\}, \{d\}, \{a, c\} \}$$

$$P_{15} = \{ \{a, b, c, d\} \}$$

The following table describes the no of relations equivalent relations and partitions of the given finite set.

| Set (A)              | No of Relations on A | No of Equivalence Relations on A | No of partitions on A. |
|----------------------|----------------------|----------------------------------|------------------------|
| $A = \{a, b\}$       | $2^2$                | 2                                | 2                      |
| $A = \{a, b, c\}$    | $2^3$                | 5                                | 5                      |
| $A = \{a, b, c, d\}$ | $2^4$                | 15                               | 15                     |

(b) No of equivalence relations = No. of partitions of A

The given set is

$Z = \{0, \pm 1, \pm 2, \dots\}$  and the relation is

$$a R b \Leftrightarrow (a, b) \text{ is divisible by } 5$$

$$\Leftrightarrow 5 \mid (a, b)$$

→ ①

$$\forall a, b \in Z$$

(i) R is Reflexive

Since  $5 \mid 0$

$$(i) \Rightarrow 5 \mid (a-a) \quad \forall a \in \mathbb{Z}$$

(18)

$$aRa, \forall a \in \mathbb{Z} \text{ by } (i)$$

(ii) R is Symmetric

$$aRb \Rightarrow 5 \mid (a-b) \text{ by } (i)$$

$$\Rightarrow 5 \mid (b-a)$$

$$\Rightarrow 5 \mid -(b-a)$$

$$\Rightarrow 5 \mid (b-a)$$

$$\Rightarrow bRa \text{ by } (i)$$

(iii) R is transitive

$$aRb \text{ and } bRc$$

$$\Rightarrow 5 \mid (a-b) \text{ and } 5 \mid (b-c) \text{ by } (i)$$

$$\Rightarrow 5 \mid (a-b) + (b-c)$$

$$\Rightarrow 5 \mid (a-c)$$

$\Rightarrow aRc$ . R is transitive.

$\therefore$  The relation (i) is equivalence relation on  $\mathbb{Z}$ .

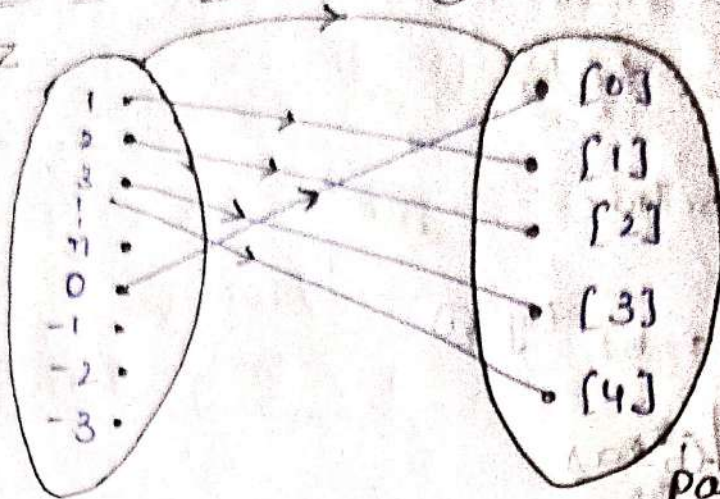
# Equivalence classes

⑨

$\mathbb{Z}$

①

Class of  $z$



partitions

$$[0] = \bar{0} = \{a \in \mathbb{Z} : a \equiv 0\}$$

$$= \{a \in \mathbb{Z} : 5 | (a - 0)\}$$

$$= \{a \in \mathbb{Z} : a - 0 = 5q\}$$

$$= \{5q + 0 : q \in \mathbb{Z}\}$$

$$[1] = \{5q + 1 : q \in \mathbb{Z}\}$$

$$[2] = \{5q + 2 : q \in \mathbb{Z}\}$$

$$[3] = \{5q + 3 : q \in \mathbb{Z}\}$$

$$[4] = \{5q + 4 : q \in \mathbb{Z}\}$$

$$[5] = [0], [6] = [1], [7] = [2], [8] = [3],$$

$$[9] = [4], [10] = [0]$$

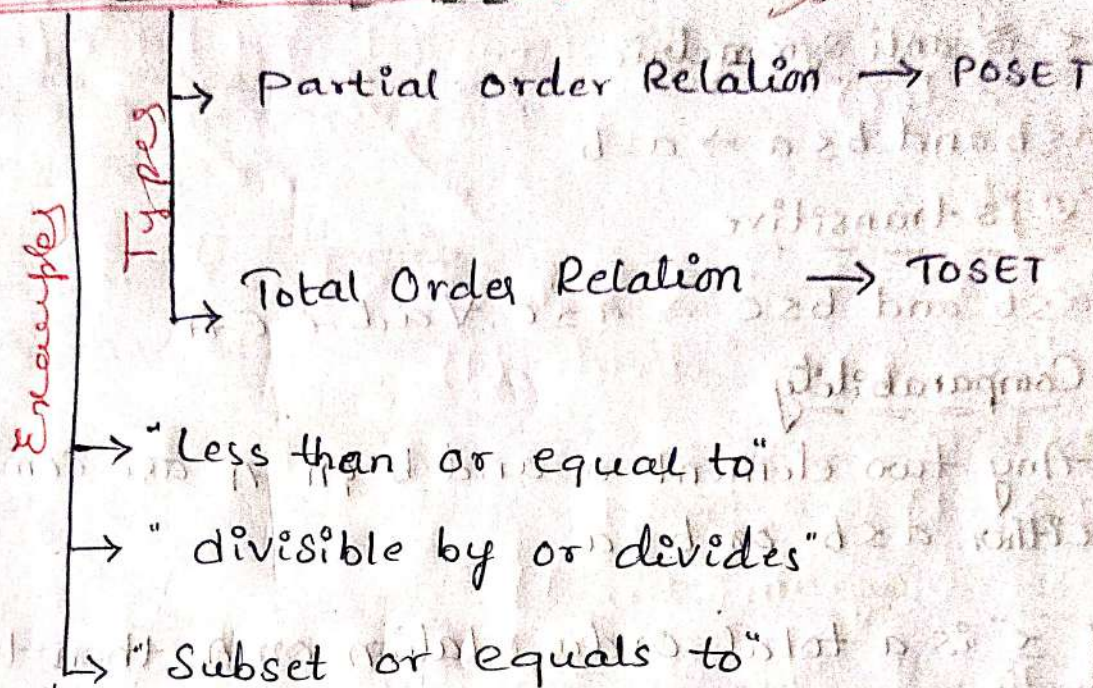
$$\therefore P = \{[0], [1], [2], [3], [4]\} = \mathbb{Z}_{15}$$

~~the relation is an equivalence relation~~

on  $\mathbb{Z}$

7. Explain order relations and define POSET and TOSET with examples, and verify whether the divisibility relation on  $\mathbb{Z}^+$  by  $nRm \Leftrightarrow n|m$  is either a partial order, a total order or both.

### order Relations of a set:



### Partial Order relation and POSET

A relation " $\leq$ " (Replace  $R$  by  $\leq$ ) is called partial order relation on the given set  $A$  if " $\leq$ " satisfies.

(i) " $\leq$ " is reflexive

$$a \leq a, \forall a \in A \quad [\because aRa, \forall a \in A]$$

(ii) " $\leq$ " is anti-symmetric

$$a \leq b \text{ and } b \leq a \Rightarrow a = b \quad [ \because aRb \Rightarrow bRa ]$$

(iii) " $\leq$ " is Transitive

$$a \leq b \text{ and } b \leq c \Rightarrow a \leq c, \forall a, b, c \in A.$$

If " $\leq$ " is a partial order relation on  $A$ , then the structure  $(A, \leq)$  is called POSET

## Total Order Relation and TOSET

(21)

A Relation " $\leq$ " (Replace  $R$  by  $\leq$ ) is called total order relation on the given Set  $A$  if  $\leq$  satisfies

(i)  $\leq$  is reflexive

$$a \leq a, \forall a \in A$$

(ii) " $\leq$ " is anti symmetric

$$a \leq b \text{ and } b \leq a \Rightarrow a = b$$

(iii) " $\leq$ " is transitive

$$a \leq b \text{ and } b \leq c \Rightarrow a \leq c, \forall a, b, c \in A.$$

(iv) Comparability

Any two element  $a$  and  $b$  on  $A$  are comparable if either  $a \leq b$  or  $b \leq a$ .

If " $\leq$ " is a total order relation on  $A$ , then the Structure  $(A, \leq)$  is called TOSET.

### Example:-

Taking  $n=6$

Set  $D_6 = \{1, 2, 3, 6\} \Rightarrow$  Divisors of 6

Relation " $\leq$ " is "divides ( $|$ )".

Structure  $\div (D_6, |)$

(i) " $|$ " is reflexive

$$[1|1, 2|2, 3|3, 6|6] \Rightarrow a|a, \forall a \in D_6 \Rightarrow "| \text{ is reflexive}$$

(ii) " $|$ " is anti-symmetric

$$a|b \text{ and } b|a \Rightarrow a = b$$

(iii) " $|$ " is transitive

$$a|b, b|c \Rightarrow a|c, \forall a, b, c \in D_6$$

(iv) Comparability

Taking  $a=2, b=3$

$2|3$  and also  $3|2$

$\therefore$  The Structure  $(D_6, |)$  is a poset, But not TOSET (22)

b The given set is  $N = \mathbb{Z}^+ = \{1, 2, 3, \dots\}$  and the relation is

$$nRm \Leftrightarrow n|m \rightarrow \textcircled{1}$$

(i) "I" is reflexive

$$a|a, \forall a \in N$$

(ii) "I" is anti-symmetric

$$a|b \text{ and } b|a \Rightarrow a=b$$

(iii) "I" is transitive

$$a|b, b|c \Rightarrow a|c, \forall a, b, c \in N$$

(iv) "I" is Comparability

Suppose  $a \neq b$  in  $N$ , then neither  $a|b$  and  $b|a$ .  
For example  $2 \nmid 3$  and also  $3 \nmid 2$  in  $N$ .

$\therefore$  The Structure  $(N, |)$  is a poset but not TOSET.

8. (8) Explain the Concept of a Hasse diagram of a poset, and Construct the Hasse diagrams for the posets (a)  $(P(\{a, b\}), \subseteq)$ , (b)  $(P(\{a, b, c\}), \subseteq)$ , and (c)  $(P(\{a, b, c, d\}), \subseteq)$ .

Hasse Diagram of a poset

Let  $(A, \leq)$  be any poset then the directed graphical representation of  $(A, \leq)$  is called a Hasse Diagram

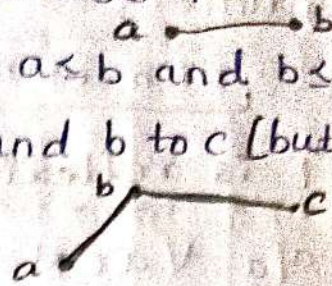
The following steps are the algorithmic procedure for Hasse diagram of a poset.

Step (1): Consider all elements in the poset  $(A, \leq)$  as Vertices (nodes) of its Hasse diagram.

Step (2): Suppose  $a \in A$  and let  $a \leq a$  then there is no edge from a to a (23)

Step (3): Suppose  $a, b \in A$  and let  $a \leq b$ . Then there is an edge from a to b (but not from b to a)

Step (4): Suppose  $a, b, c \in A$  and let  $a \leq b$  and  $b \leq c$ . Then there is an edge from a to b and b to c (but not a to c)



Example:

$D_{12} = \{1, 2, 3, 4, 6, 12\}$ , Divisors of 12

Relation: " $\leq$ " is "divides (|)".

Structure:

$(D_{12}, |)$  is a POSET

Hasse Diagram  $(D_{12}, |)$

$1/1, 2/2, 3/3, 4/4, 6/6, 12/12$  Step - 2

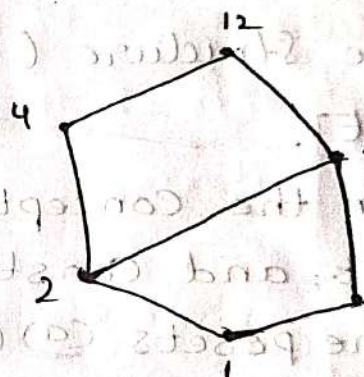
$1/2, 1/3, 1/4, 1/6, 1/12$

$2/3, 2/4, 2/6, 2/12$

$3/4, 3/6, 3/12$

$4/6, 4/12$

$6/12$



Q1) Set  $S = \{a, b\}$

power set

$P(S) = \{\emptyset, \{a\}, \{b\}, S\}$

Relation

The relation " $\leq$ " is Subset or equal to ( $\subseteq$ )

Structure

$(P(S), \subseteq)$  is a POSET. (TOSET)

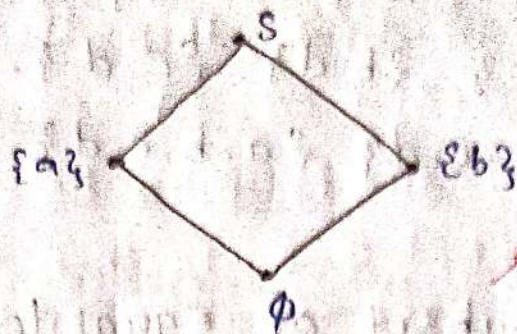
Hasse - diagram

$\emptyset \subseteq \emptyset, \{a\} \subseteq \{a\}, \{b\} \subseteq \{b\}, S \subseteq S$

$\emptyset \subseteq \{a\}, \emptyset \subseteq \{b\}, \emptyset \subseteq S$

$$\{a\} \not\subseteq \{b\}, \{a\} \subseteq S$$

$$\{b\} \not\subseteq \{a\}, \{b\} \subseteq S$$



(iv) Set  $S = \{a, b, c\}$ .

$$P(S) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, S\}$$

Relation

The relation " $\subseteq$ " is "Subset or equal to ( $\subseteq$ )"

Structure

$(P(S), \subseteq)$  is a POSET, CTOSET.

Hasse-diagram.

$$\phi \subseteq \{a\}, \phi \subseteq \{b\}, \phi \subseteq \{c\}, \phi \subseteq \{a, b\}, \phi \subseteq \{a, c\},$$

$$\phi \subseteq \{b, c\}, \phi \subseteq S$$

$$\{a\} \not\subseteq \{b\}, \{a\} \not\subseteq \{c\}, \{a\} \subseteq \{a, b\}, \{a\} \subseteq \{a, c\}, \{a\} \not\subseteq \{b, c\}, \{a\} \subseteq S,$$

$$\{b\} \not\subseteq \{c\}, \{b\} \subseteq \{a, b\}, \{b\} \not\subseteq \{a, c\}, \{b\} \subseteq \{b, c\}, \{b\} \subseteq S,$$

$$\{c\} \not\subseteq \{a, b\}, \{c\} \subseteq \{a, c\}, \{c\} \subseteq \{b, c\}, \{c\} \subseteq S$$

$$\{a, b\} \not\subseteq \{a, c\}, \{a, b\} \not\subseteq \{b, c\}, \{a, b\} \subseteq S$$

$$\{a, c\} \not\subseteq \{b, c\}, \{a, c\} \subseteq S$$

$$\{b, c\} \subseteq S$$

$$S = \{a, b, c\}$$

$$\{a, b\}$$

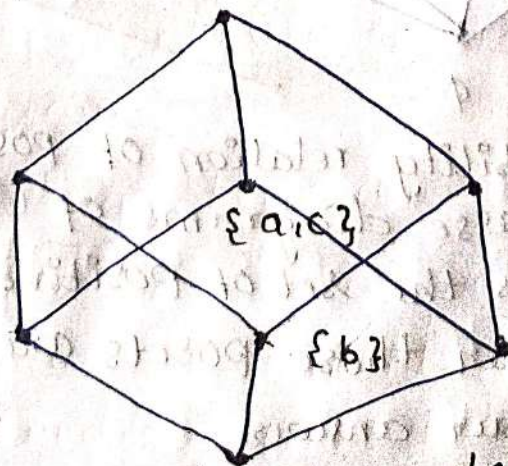
$$\{a, c\}$$

$$\{b, c\}$$

$$\{a\}$$

$$\{b\}$$

$$\{c\}$$



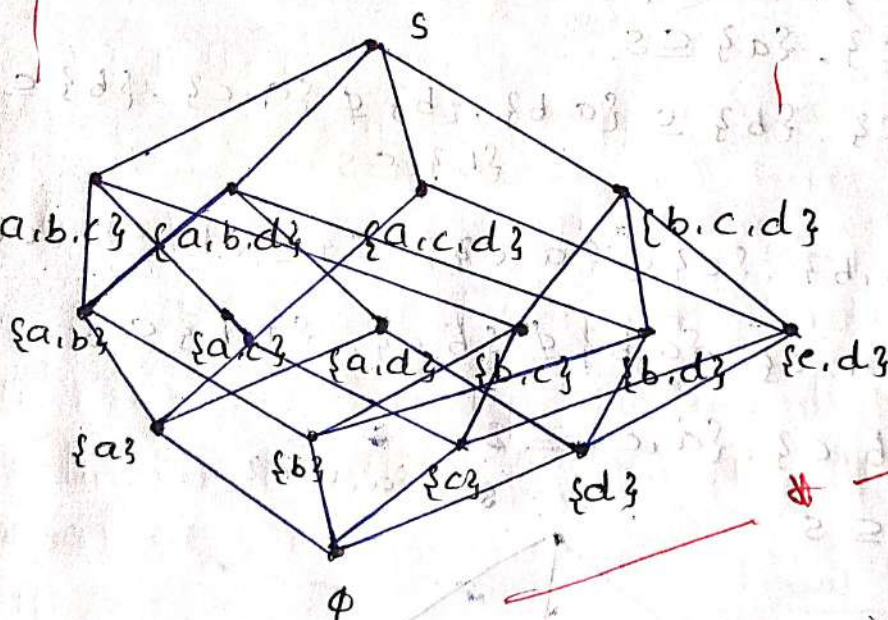
(iii)

Set  $S = \{a, b, c, d\}$ 

$$P(S) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{b, c, d\}, \{a, c, d\}, \{a, b, c, d\}, S\}$$

RelationThe relation " $\subseteq$ " is "Subset or equal to ( $\subseteq$ )".Structure $(P(S), \subseteq)$  is a POSET. (TOSET)Hasse diagram

$$\begin{aligned} \emptyset &\subseteq \{a\}, \emptyset \subseteq \{b\}, \emptyset \subseteq \{c\}, \emptyset \subseteq \{d\}, \emptyset \subseteq \{a, b\}, \emptyset \subseteq \{a, c\}, \emptyset \subseteq \{a, d\}, \emptyset \subseteq \{b, c\}, \emptyset \subseteq \{b, d\}, \emptyset \subseteq \{c, d\} \\ \emptyset &\subseteq S \end{aligned}$$



9. Use the divisibility relation of positive integers to construct the Hasse diagrams of  $(D_{12}, |)$  and  $(D_{24}, |)$ , where  $D_n$  denotes the set of positive divisors of  $n$ . Determine whether those posets are total orders and justify your answer.

$(D_{12}, 1)$  :

$$D_{12} = \{1, 2, 3, 4, 6, 12\}$$

Relation: " $\mid$ " is divides

Structure

$(D_{12}, 1)$  is a POSET

Hasse Diagram  $(D_{12}, 1)$

$$1/1, 2/2, 3/3, 4/4, 6/6, 12/12$$

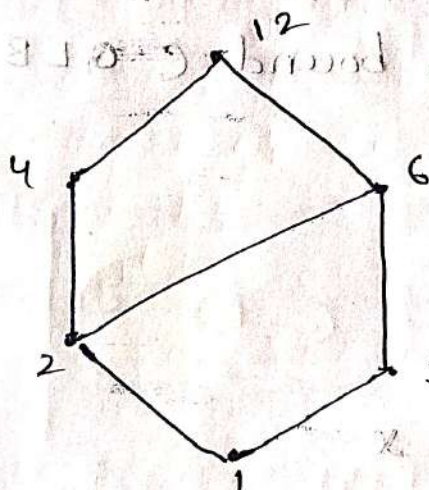
$$1/2, 1/3, 1/4, 1/6, 1/12$$

$$2/3, 2/4, 2/6, 2/12$$

$$3/4, 3/6, 3/12$$

$$4/6, 4/12$$

$$6/12$$



It is a POSET but not TOSET because  $2 \nmid 3$  and  $3 \nmid 2$  (i.e. 2 and 3 are not comparable)

Supremum and Infimum

$(D_{24}, 1)$  :

$$D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$$

Relation: " $\mid$ " is divides

Structure

$(D_{24}, 1)$  is POSET

Hasse Diagram

$$1/1, 2/2, 3/3, 4/4, 6/6, 8/8, 12/12, 24/24$$

$$1/2, 1/3, 1/4, 1/6, 1/8, 1/12, 1/24$$

$$2/3, 2/4, 2/6, 2/8, 2/12, 2/24$$

$$3/4, 3/6, 3/8, 3/12, 3/24$$

$$4/6, 4/8, 4/12, 4/24$$

$$6/8, 6/12, 6/24$$

$$8/12, 8/24$$

$$12/24$$

$$24/24$$

$$24/24$$

$$24/24$$

$$24/24$$

$$24/24$$

$$24/24$$

$$24/24$$

$$24/24$$

$$24/24$$

$$24/24$$

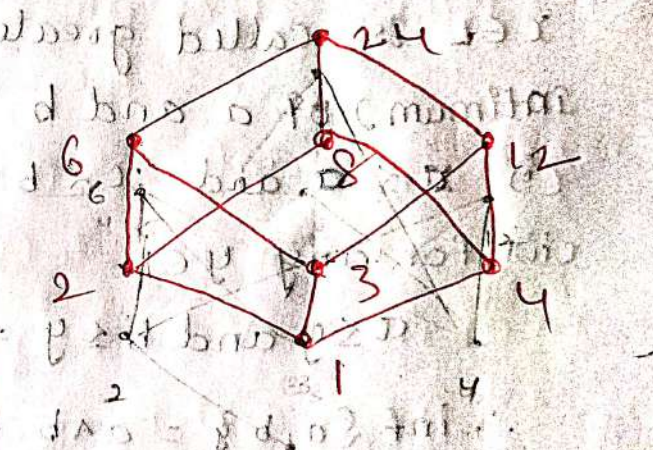
$$24/24$$

$$24/24$$

$$24/24$$

$$24/24$$

$$24/24$$



It is also a POSET but not TOSET because  $2 \nmid 3$  and  $3 \nmid 2$  (i.e. 2 and 3 are not comparable in  $D_{24}$ )

## Supremum and Infimum

(22)

Let  $L$  be any non-empty set and let " $\leq$ " be a partial (Total) order on  $L$ . Then there exists a POSET (or) TOSET  $(L, \leq)$ .

An element  $x \in L$  is called upper bound of  $L$  if  $a \leq x$  for all  $a \in L$ . Let  $a, b \in L$ , then an element  $x \in L$  is called least upper bound (LUB or Supremum) if  $a$  and  $b$  if

- (i)  $a \leq x$  and  $b \leq x$
- (ii) for any  $y \in L$ ,

$$a \leq y \text{ and } b \leq y \Rightarrow x \leq y$$

$$\therefore \sup \{a, b\} = a \vee b = a \oplus b = x$$

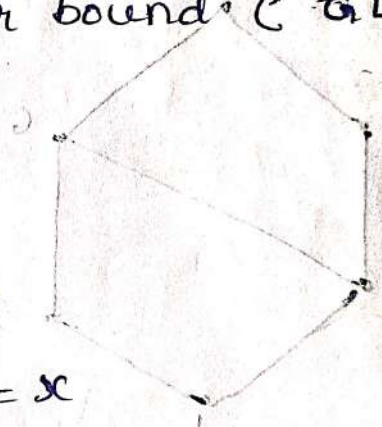
An element  $x \in L$  is called lower bound of  $L$  if  $x \leq a$  for all  $a \in L$ . Let  $a, b \in L$ , then an element  $x \in L$  is called greatest lower bound (GLB or Infimum) of  $a$  and  $b$  if

(i)  $x \leq a$  and  $x \leq b$

(ii) for any  $y \in L$

$$a \leq y \text{ and } b \leq y \Rightarrow x \leq y$$

$$\therefore \inf \{a, b\} = a \wedge b = a * b = x$$



10. Describe lattice and explain its properties with examples. Verify whether the following structures are lattices: (a)  $(\mathbb{N}, |)$ , (b)  $(\mathbb{Z}, \leq)$  and (c)  $(P(S), \subseteq)$ .

Sol: Lattice ✓

The structure  $(L, \leq)$  is called a Lattice if

- (i)  $(L, \leq)$  is a POSET (TOSET)

(ii) " $\leq$ " is reflexive

$$a \leq a, \forall a \in L$$

(i) " $\leq$ " is anti-symmetric

$$a \leq b \text{ and } b \leq a \Rightarrow a = b \text{ in } L$$

(28)

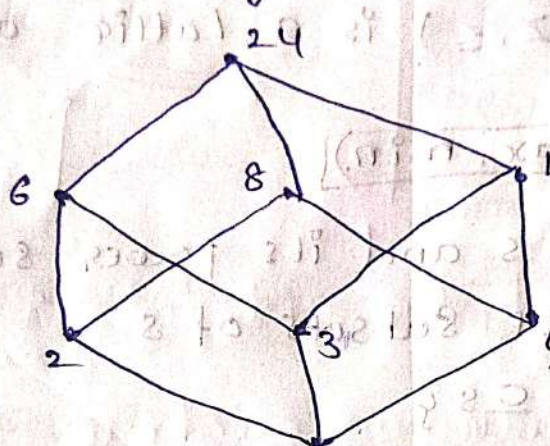
(ii) " $\leq$ " is transitive

$$a \leq b \text{ and } b \leq c \Rightarrow a \leq c \text{ in } L$$

(2) Existence of sup and inf

For any  $a, b \in L$ , there exists  $\sup\{a, b\} \in L$   
and  $\inf\{a, b\} \in L$ .

Ex: The structure  $(D_{24}, |)$  is a lattice with  
the hasse diagram.



(a) the given set is  $N = \{1, 2, 3, \dots\}$  and the  
relation "is divisible by" ( $a|b$ )

(i)  $a|a, \forall a \in N$ .

(ii)  $a|b \text{ and } b|a \Rightarrow a = b \text{ in } N$

(iii)  $a|b \text{ and } b|c \Rightarrow a|c \text{ in } N$

$\therefore$  The structure  $(N, |)$  is a POSET.

Existence of sup and inf

$$\sup\{a, b\} = a \vee b = a \oplus b = \text{LCM}\{a, b\} \in N$$

$$\inf\{a, b\} = a \wedge b = a \div b = \text{gcd}\{a, b\} \in N$$

hence, the structure  $(N, |)$  is a lattice,

and it is denoted by  $[N, \text{LCM}, \text{gcd}]$

(if) The given set is  $Z = \{0, \pm 1, \pm 2, \dots\}$  and the relation "is less than or equal to" ( $\leq$ ) (29)

(i)  $a \leq b, \forall a \in Z$

(ii)  $a \leq b$  and  $b \leq a \Rightarrow a = b$  in  $Z$

(iii)  $a \leq b$  and  $b \leq c \Rightarrow a \leq c$  in  $Z$

$(Z, \leq)$  is a POSET

(iv) Existence of Sup and Inf

$\text{Sup}\{a, b\} = a \vee b = a \oplus b = \max\{a, b\} \in Z$

$\text{Inf}\{a, b\} = a \wedge b = a * b = \min\{a, b\} \in Z$

$\therefore$  The Structure  $(Z, \leq)$  is a Lattice and it is also denoted by

$(Z, \max, \min)$

3. (c) The given set is  $S$  and its power set is

$P(S) = \text{Set of all Subsets of } S$

$= \{A : A \subseteq S\}$

and the relation  $P(S)$  is

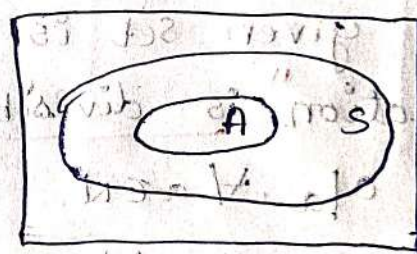
"Subset or equal to" ( $\subseteq$ )

(i) (i)  $A \subseteq A, \forall A \in P(S)$

(ii)  $A \subseteq B$  and  $B \subseteq A \Rightarrow A = B$  in  $P(S)$

(iii)  $A \subseteq B$  and  $B \subseteq C \Rightarrow A \subseteq C$  in  $P(S)$

$\therefore (P(S), \subseteq)$  is a POSET



(2) Existence of Sup and Inf

$\text{Sup}\{A, B\} = A \vee B = A \oplus B = A \cup B \in P(S)$

$\text{Inf}\{A, B\} = A \wedge B = A * B = A \cap B \in P(S)$

$\therefore$  The Structure  $(P(S), \subseteq)$  is a lattice and it is also denoted by  $(P(S), \cup, \cap)$

11. Construct the Hasse diagrams of the posets  $(P_{30}, \subseteq)$  and  $(P\{1, 2, 3\}, \subseteq)$  and Verify whether these posets form lattices or not.

Sol:

Taking  $n = 30$

Divisors of 30

$$D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$$

Relation on  $D_{30}$

"is divisible by (1)"

Structure

$(D_{30}, |)$

(1)  $(D_{30}, |)$  is a POSET

(2) Existence of Sup & Inf

$$\text{Sup } \{a, b\} = \text{Lcm } \{a, b\}$$

| Lcm | 1  | 2  | 3  | 5  | 6  | 10 | 15 | 30 |
|-----|----|----|----|----|----|----|----|----|
| 1   | 1  | 2  | 3  | 5  | 6  | 10 | 15 | 30 |
| 2   | 2  | 2  | 6  | 10 | 6  | 10 | 30 | 30 |
| 3   | 3  | 6  | 3  | 15 | 30 | 30 | 15 | 30 |
| 5   | 5  | 10 | 15 | 5  | 6  | 10 | 15 | 30 |
| 6   | 6  | 6  | 6  | 30 | 6  | 30 | 30 | 30 |
| 10  | 10 | 10 | 30 | 10 | 30 | 10 | 30 | 30 |
| 15  | 15 | 30 | 15 | 15 | 30 | 30 | 15 | 30 |
| 30  | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 |

$$\text{inf } \{a, b\} = \text{gcd } \{a, b\}$$

| gcd | 1 | 2 | 3 | 5 | 6 | 10 | 15 | 30 |
|-----|---|---|---|---|---|----|----|----|
| 1   | 1 | 1 | 1 | 1 | 1 | 1  | 1  | 1  |
| 2   | 1 | 2 | 1 | 1 | 2 | 2  | 1  | 2  |
| 3   | 1 | 1 | 3 | 1 | 3 | 1  | 3  | 3  |
| 5   | 1 | 1 | 1 | 5 | 1 | 5  | 5  | 5  |
| 6   | 1 | 2 | 3 | 1 | 6 | 2  | 3  | 6  |
| 10  | 1 | 2 | 1 | 5 | 2 | 10 | 5  | 10 |
| 15  | 1 | 2 | 3 | 5 | 3 | 5  | 15 | 15 |
| 30  | 1 | 2 | 3 | 5 | 6 | 10 | 15 | 30 |

Given

Set  $S = \{1, 2, 3\}$

Power Set

$$P(S) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$$

$$= \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$$

Structure

$(P(S), \subseteq)$

(i)  $A \subseteq A, \forall A \in P(S)$

(ii)  $A \subseteq B$  and  $B \subseteq A \Rightarrow A = B$  in  $P(S)$

(iii)  $A \subseteq B$  and  $B \subseteq C \Rightarrow A \subseteq C$  in  $P(S)$

$(P(S), \subseteq)$  is a POSET.

2. Existence of Sup and Inf

| $u$         | $\emptyset$ | $\{1\}$    | $\{2\}$    | $\{3\}$    | $\{1, 2\}$ | $\{1, 3\}$ | $\{2, 3\}$ | $S$ |
|-------------|-------------|------------|------------|------------|------------|------------|------------|-----|
| $\emptyset$ | $\emptyset$ | $\{1\}$    | $\{2\}$    | $\{3\}$    | $\{1, 2\}$ | $\{1, 3\}$ | $\{2, 3\}$ | $S$ |
| $\{1\}$     | $\{1\}$     | 1          | $\{1, 2\}$ | $\{1, 3\}$ | $\{1, 2\}$ | $\{1, 3\}$ | $\{2, 3\}$ | $S$ |
| $\{2\}$     | $\{2\}$     | $\{2\}$    | $\{2\}$    | $\{2, 3\}$ | $\{1, 2\}$ | $\{3\}$    | $\{2, 3\}$ | $S$ |
| $\{3\}$     | $\{3\}$     | $\{3\}$    | $\{3\}$    | $\{3\}$    | $\{3\}$    | $\{1, 3\}$ | $\{2, 3\}$ | $S$ |
| $\{1, 2\}$  | $\{1, 2\}$  | $\{1, 2\}$ | $\{1, 2\}$ | $\{1, 2\}$ | $\{1, 2\}$ | $\{1, 2\}$ | $\{2, 3\}$ | $S$ |
| $\{1, 3\}$  | $\{1, 3\}$  | $\{1, 3\}$ | $\{1, 3\}$ | $\{1, 3\}$ | $\{1, 3\}$ | $\{1, 3\}$ | $\{2, 3\}$ | $S$ |
| $\{2, 3\}$  | $\{2, 3\}$  | $\{2, 3\}$ | $\{2, 3\}$ | $\{2, 3\}$ | $\{2, 3\}$ | $\{2, 3\}$ | $\{2, 3\}$ | $S$ |
| $S$         | $S$         | $S$        | $S$        | $S$        | $S$        | $S$        | $S$        | $S$ |

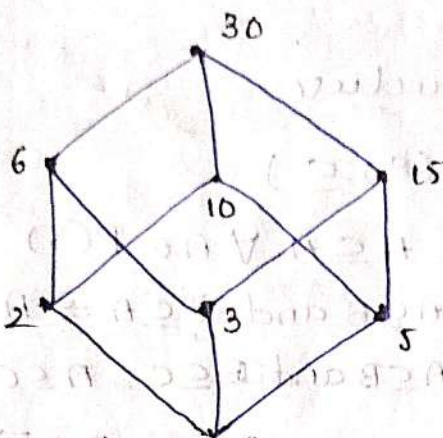
| $n$         | $\emptyset$ | $\{1\}$     | $\{2\}$     | $\{3\}$     | $\{1, 2\}$  | $\{1, 3\}$  | $\{2, 3\}$  | $S$         |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| $\emptyset$ | $\emptyset$ | $\emptyset$ | $\emptyset$ | $\emptyset$ | $\emptyset$ | $\emptyset$ | $\emptyset$ | $\emptyset$ |
| $\{1\}$     | $\emptyset$ | $\{1\}$     | $\emptyset$ | $\emptyset$ | $\{1\}$     | $\{1\}$     | $\emptyset$ | $\{1\}$     |
| $\{2\}$     | $\emptyset$ | $\emptyset$ | $\{2\}$     | $\emptyset$ | $\{2\}$     | $\emptyset$ | $\{2\}$     | $\{2\}$     |
| $\{3\}$     | $\emptyset$ | $\emptyset$ | $\emptyset$ | $\{3\}$     | $\emptyset$ | $\{3\}$     | $\{3\}$     | $\{3\}$     |
| $\{1, 2\}$  | $\emptyset$ | $\{1\}$     | $\{2\}$     | $\emptyset$ | $\{1, 2\}$  | $\{1\}$     | $\{2\}$     | $\{1, 2\}$  |
| $\{1, 3\}$  | $\emptyset$ | $\{1\}$     | $\emptyset$ | $\{3\}$     | $\emptyset$ | 1           | $\{3\}$     | $\{1, 3\}$  |
| $\{2, 3\}$  | $\emptyset$ | $\emptyset$ | $\{2\}$     | $\emptyset$ | $\emptyset$ | $\{2\}$     | $\{2, 3\}$  | $\{2, 3\}$  |
| $S$         | $\emptyset$ | 1           | 2           | 3           | $\{1, 2\}$  | $\{1, 3\}$  | $\{2, 3\}$  | $S$         |

From the above table,

$\text{Sup}\{a, b\}$  and  $\text{Inf}\{a, b\} \in D_{30}$

Hence,  $(D_{30}, \leq)$  is a lattice and it is also denoted by  $(D_{30}, \cup, \cap)$

Hasse diagram



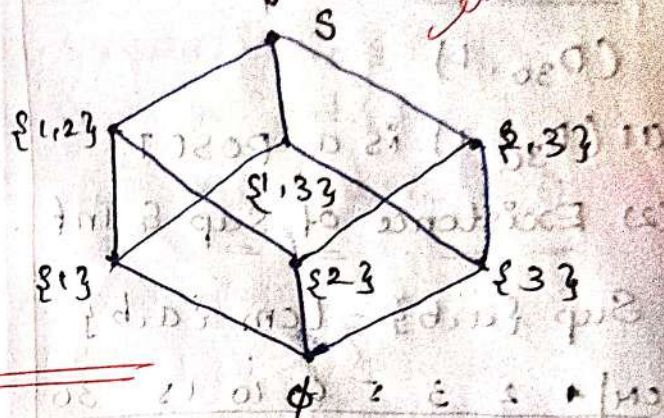
From the above table

$\text{Sup}\{A, B\} = A \vee B = A \oplus B = A \cup B \in P(S)$

$\text{Inf}\{A, B\} = A \wedge B = A * B = A \cap B \in P(S)$

$\therefore$  The structure  $(P(S), \leq)$  is Lattice, and it is also denoted by  $(P(S), \cup, \cap)$

Hasse diagram



⑫ Explain Boolean algebra and list its fundamental properties.

Verify that the following algebraic structures are Boolean algebras:

(a)  $(P(S), \cup, \cap, \emptyset, S)$

(b)  $(D_{30}, \oplus, *, 1, \pm_{30})$

(c)  $(D_{24}, \oplus, *, 1, \pm_{24})$

Sol: (a) Boolean algebra

Let  $B$  be any finite set and let " $\leq$ " be the relation on  $B$ . Then the structure  $(B, \leq)$  is called a Boolean Algebra if it satisfies the following:

(1)  $(B, \leq)$  is a POSET (TOSET);

(2) Existence Sup and Inf

For any  $a, b \in B$  ( $a \neq b$ ), then exists,  $\text{Sup}\{a, b\}$ ,  $\text{Inf}\{a, b\}$  such that  $\text{Sup}\{a, b\} \in B$  and  $\text{Inf}\{a, b\} \in B$

(3) Existence of Complement

$\therefore$  for any  $a \in B$ , there exists  $a'$  ( $\text{or } \bar{a}$ ) in  $B$  such that  $a'$  is called Complement  $a$  in  $B$ .

#### 4. Existence of Bounds (Identities)

The elements 0 and 1 are called identities of B  
Such that  $a \oplus a' = 1$ ,  $a * a' = 0$ .

The Boolean structure  $(B, \leq)$  is also denoted by

$$(B, \oplus, *, ', 0, 1)$$

Properties:

##### 1. Commutative property

$$\left. \begin{aligned} a \oplus b &= b \oplus a \\ a * b &= b * a \end{aligned} \right\}$$

##### 2. Associative properties

$$\left. \begin{aligned} a \oplus b \oplus c &= (a \oplus b) \oplus c \\ a * b * c &= (a * b) * c \end{aligned} \right\}$$

##### 3. Distributive properties

$$a * (b \oplus c) = (a * b) \oplus (a * c)$$

$$a \oplus (b * c) = (a \oplus b) * (a \oplus c)$$

##### 4. Absorption laws

$$\left. \begin{aligned} a \oplus (a * b) &= a \\ a * (a \oplus b) &= a \end{aligned} \right\}$$

##### 5. Identity laws

$$a \oplus 0 = a = 0 \oplus a, \forall a \in B$$

$$a * 1 = a = 1 * a, \forall a \in B$$

(a) The given set is S and  
its power set is

$$P(S) = \{A : A \subseteq S\}$$

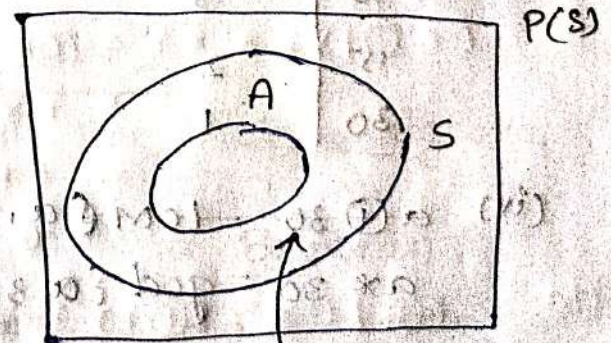
Structure:  $(P(S), \subseteq)$ .

(i)  $(P(S), \subseteq)$  is a poset

(ii)  $\sup \{A, B\} = A \oplus B = A \cup B \in P(S)$

(iii) For  $A \in P(S)$  there exists

$A' \in P(S)$  such that  $A' = S - A$



$$(iv) A \oplus \phi = A \cup \phi = A$$

$$A + S = A \cap S = A$$

$\Rightarrow \phi$  and  $S$  are identities, with

$$\left. \begin{aligned} A \oplus A' &= A \cup A' = S \\ A * A' &= A \cap A' = \phi \end{aligned} \right\}$$

$\therefore$  The Structure  $(P(S), \cup, \cap, 1, \phi, S)$  is a Boolean Algebra.

(b) The given Set is

$$D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$$

Structure:  $(D_{30}, 1)$

(i)  $(D_{30}, 1)$  is a POSET

(ii)  $\text{Sup}\{a, b\} = a \oplus b = \text{LCM}\{a, b\} \in D_{30}$

(iii)  $\text{Inf}\{a, b\} = a * b = \text{gcd}\{a, b\} \in D_{30}$

(iv) For any  $a \in D_{30}$ , there exists  $a'$  in  $D_{30}$  such that

$$a' = \frac{30}{a}$$

| $a$ | $a' = \frac{30}{a}$ |
|-----|---------------------|
| 1   | 30                  |
| 2   | 15                  |
| 3   | 10                  |
| 4   |                     |
| 5   | 6                   |
| 6   | 5                   |
| 10  | 3                   |
| 15  | 2                   |
| 30  | 1                   |

(iv)  $a \oplus 30 = \text{LCM}\{a, 30\} \Rightarrow a \oplus 1 = \text{LCM}\{a, 1\} = a$

$$a * 30 = \text{gcd}\{a, 30\} = a'$$

$\therefore 1$  &  $30$  are identity elements in  $(D_{30})$  (with (i))

$$a \oplus a' = \text{LCM}\{a, a'\} = 30$$

$$a * a' = \text{gcd}\{a, a'\} = 1$$

$\therefore$  The Structure  $(D_{30}, \text{LCM}, \text{gcd}, 1, 1, 30)$  is a Boolean algebra

(C) The given Set is

$$D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$$

Structure :-  $(D_{24}, 1)$

(i)  $(D_{24}, 1)$  is a poset

(ii)  $\text{Sup} \{a, b\} = a \oplus b \rightarrow \text{LCM} \{a, b\} \in D_{24}$

$\text{Inf} \{a, b\} = a * b \rightarrow \text{gcd} \{a, b\} \in D_{24}$

(iii) For any  $a \in D_{24}$ , there exists  $a'$  in  $D_{24}$  such that  $a' = \frac{24}{a}$

| a  | $a' = \frac{24}{a}$ |
|----|---------------------|
| 1  | 24                  |
| 2  | 12                  |
| 3  | 8                   |
| 4  | 6                   |
| 6  | 4                   |
| 8  | 3                   |
| 12 | 2                   |
| 24 | 1                   |

(iv)  $a \oplus 1 = \text{LCM} \{a, 1\} = a$   
 $a * 24 = \text{gcd} \{a, 24\} = a$

$\therefore 1$  and  $24$  are identity elements in  $D_{24}$  with.

$a \oplus a' = \text{LCM} \{a, a'\} \neq 24$  because  $4 \oplus 6 = \text{LCM} \{4, 6\} = 12 \neq 24$

$a * a' = \text{gcd} \{a, a'\}$

Hence, the Structure

$$(D_{24}, \text{LCM}, \text{gcd}, (1, 24))$$

is a poset and lattice but not boolean algebra.

⑬ Show that  $(B_2, \oplus, *, \square, 0, 1)$  is a Boolean algebra, and use its properties to evaluate

(a)  $(0 \oplus 1) * 0 * 1$ ,  $0 \oplus 0$

(b)  $0 \oplus [0 * (1 \oplus 1)]$

(c)  $(1 * 0) \oplus (1 * 0) * (0 \oplus 1)$ , and

(d)  $(0 \oplus 1 * 0) * (1 * 0 \oplus 1)$ .

The given set is  $B_2 = \{0, 1\}$

Where  $0^2 = 0, 1^2 = 1$

Define  $\oplus, *$  and  $'$  on  $B_2$  as

| $\oplus$ | 0 | 1 |
|----------|---|---|
| 0        | 0 | 1 |
| 1        | 1 | 1 |

| $*$ | 0 | 1 |
|-----|---|---|
| 0   | 0 | 0 |
| 1   | 0 | 1 |

| $a$ | $a'$ |
|-----|------|
| 0   | 1    |
| 1   | 0    |

(1)  $B_2$  is a POSET

(2) Existence of Sup and Inf

$$\text{Sup}\{0, 1\} = 0 \oplus 1 = 1 \in B_2$$

$$\text{Inf}\{0, 1\} = 0 * 1 = 0 \in B_2$$

(3) Existence of Complement

$$0' = 1 \in B_2 \text{ and}$$

$$1' = 0 \in B_2$$

(4) Bounds (Identities) of  $B_2$

Since  $0, 1 \in B_2$

$$\therefore a \oplus 0 = a = 0 \oplus a, \forall a \in B_2$$

$$a * 1 = a = 1 * a, \forall a \in B_2$$

with

$$a \oplus a' = 1 = a' \oplus a$$

$$a * a' = 0 = a' * a$$

$\therefore$  The Structure

$(B_2, \oplus, *, ', 0, 1)$  is a boolean algebra.

(a)  $(0 \oplus 1) * 0' * 1' = (0 \oplus 1) * 1 * 0$

$$= 1 * 1 * 0$$

$$= 1 * 0$$

$$= 0$$

(b)  $0 \oplus [0 * (1 \oplus 1)] = 0 \oplus [0 * 1]$

$$= 0 \oplus 0$$

$$= 0$$

$$(c) (1*0) \oplus (1*0) * (0 \oplus 1)$$

$$\oplus = 0 \oplus 0 * 1$$

$$\oplus = 0 * 1$$

$$\oplus = 0$$

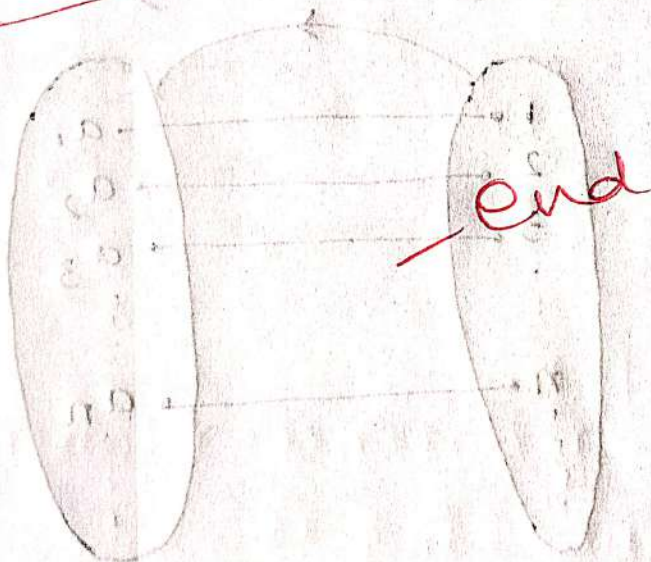
(70)

$$(d) (0 \oplus 1 * 0) * (1 * 0 \oplus 1)$$

$$\oplus = (1 * 0) * (0 \oplus 1)$$

$$\oplus = 0 * 1$$

$$\oplus = 0$$



end

the set of natural numbers and let it be the set of real numbers. Then any of fractions from it to be called "rational" numbers. It is checked.

$$[a : b] \rightarrow [a : b]$$

and represented by  $\frac{a}{b}$ . In this way, we can represent all rational numbers. The function  $f(x) = \frac{a}{b}$  is a function from the set of natural numbers to the set of rational numbers.